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# MTH401 Midterm Solved Subjective Paper

## Paper#01

## MTH 401 May 19, 2011

1) modification of logistic equation (2 marks)lec-12

$$p' = ap - bp^2$$
$$dx/dt = p(ap - b\ln p)$$

2) Auxiliary equation for DE $y'' - 4y' + 4y = 2e^{2x}$ (2 marks)

$sol:$

$$m^2 - 4m + 4 = 2e^{2x}$$

(3)harmonic motion **x(t)=c1coswt +c2sinwt** write into alternative form (3 marks)lec-22

Solution:


$\mathbf{x(t) = c1coswt\ + c2sinwt}$

Let $A, \phi \ni R$

$c_1 = A\sin\phi, c_{2=} A\cos\phi$

*then*

$A = \sqrt{c_1^{\ 2} + c_2^{\ 2}}, \tan\phi = \frac{c_1}{c_2}$

*so* that,

$x(t) = A\sin\phi\cos\omega t + A\cos\phi\sin\omega t$

$x(t) = A\sin\omega t\cos\phi + A\cos\omega t\sin\phi$

$x(t) = A\sin(\omega t + \phi)$

3) complementary solution for $y''' - 2y'' + 21y' - 18y = 3 + 4e^{-t}$ (5 marks)

$y''' - 2y'' + 21y' - 18y = 3 + 4e^{-t}$

$sol:$

$m^3 - 2m^2 + 21m - 18 = 3 + 4e^{-t}$

$m^3 - 2m^2 + 21m - 18 = 0$

$m^3 - 2m^2 + 21m = 18$

$m_1 = 1 + \sqrt{21}, m_2 = 1 - \sqrt{21}$

$y_c = c_1 e^{mx} + c_2 e^{mx}$

$y_c = c_1 e^{(1+\sqrt{21})x} + c_2 e^{(1-\sqrt{21})x}$

4) newton's cooling law (5 marks)


**paper#02**

my today's mth401 paper

1..write all the forms of g(x) in the method of undetermined coefficients where g(x) is rght hand side of Non-homogeneous linear differential equation ?

solution:

$$a_o \frac{d^n y}{dx^n} + a_n \frac{d^{n-1} y}{dx^{n-1}} + \dots\dots\dots + a_{n-1} \frac{dy}{dx} + a_n y = G(x)$$

2. Deduce the special use of logistic Equation "Epidemic Spread"?

Solution:

$$x(t) = \text{people who infected}$$
$$\text{y}(t) = \text{people who not infected}$$
$$\frac{dy}{dx} = kxy$$
$$x + y = n + 1$$
$$y = n + 1 - x$$
$$\frac{dy}{dx} = kx(n + 1 - x)$$

3.what will be the auxiliary Equation of 2nd order Homogeneous Linear Differential equation by taking solution of exponential form

$$y'' - 2y' + y = e^{t/t^2} + 1$$
$$sol:$$
$$m^2 - 2m + 1 = 0$$


4..find the auxiliary equation for the differential equation

$y'''-2y''-21y'-18y=3+4e^{-t}$

$sol:$

$m^3-2m^2-21m^1-18m^0=0$

$m^3-2m^2-21m\ \ -18=0$

5. Define general linear equation of order n ?lec 2

$$a_n(t)y(n)+a_{n-1}(t)y(n-1)+.......+a_1y'+a_o(t)y=h(t)$$

**paper#03**

My Today Paper MT H401 (Dated 04-12-2011)

Question No 1: Define cycle of a vibrating body (2)

Ans:I n travelling from x=A to x=-A,then back to A the vibrating body completes one vibration or one cycle.

Question No 2: Construct the auxiliary equation of the following differential equation:

$3d^2/dx^2-4dy/dx=0$

Solution:

$3m^2-4m=0$

Question No 3: Solve the following intial value problem

$$\int\frac{dT}{T-T_m}=\int K(dt),T(0)=T0$$


Question No 4: Factorize the differential equation D4 – 8d2 + 16 if possible

Solve:

D4-4d2-4d2+16

D2( D2-4)-4(D2-4)

(D2-4)(D2-4)

[(D)2-(2)2] [(D)2-(2)2]

(D-2) (D+2) (D-2) ( D+2)

Question No 5: Find the wronsktin of the differential equation

Y``` -2y``-18y = 6 + 4-t using variation of parameter and the root of the auxiliary equation is m1 = -3 , m2 = -1 , m3 = 6?

Question No 6: A radioactive isotope has a half-life of 16 days we have 30 g at the end of 30days, how much radio isotope was initially present?

The solution of the IVP (initial value Problem) is given by A (T) = A0 ekt , where K = Ln2/ 16

Paper#04

**MID TERM paper Of MTH 401**
**fall 2010**


***bc080400071***

***Muhammad tahir***

**Q1)** Define super position principle of particular solution non Homogeneous Linear deferential equation?

**Q2)** If a mass weighing15lb stretches a spring 3ft then fid K using hook's law?

$f = ks$

$5 = 3k$

$k = 5/3$

**Q3)** what is an auxiliary equation of a Homogeneous differential equation .also give its two examples.

Q4) what is the general solution of deferential equation?

General solution:Complementary+any particular solution

**Q5)** Take a deferential equation

□□□□

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

we take its exponential solution

$y = e^{mx}$, by putting the above solution we get equation

$e^{mx}(am^2 + bm + c) = 0$

What will

be its general solution if roots are distinct real roots?

**Q6)** Find the complementary solution for the equation □□□□ $y'' - 3y' + 2 = (x+1)e^{2x}$

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$y'' - 3y' + 2 = (x+1)e^{2x}$

$m^2 - 3m + 2 = 0$

$m^2 - m - 2m + 2 = 0$

$m(m-1) - 2(m-1) = 0$

$(m-1)(m-2) = 0$

$m = 1, m = 2$

$y_c = y_1 e^{mx} + y_2 e^{mx}$

$y_c = y_1 e^{(1)x} + y_2 e^{(2)x}$

**Paper no: 5**

**MTH401 (Dated 04-12-2011)**

**Question No 1:** Define cycle of a vibrating body (2)

Answer:

In travelling from x = A to x = - A and then back to A, the vibrating body completes one vibration or one cycle.


**Question No 2:** Construct the auxiliary equation of the following differential equation

$$3\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$$

*dividing* 1 *by equation*

$$\frac{dx^2}{3d^2y} - 4\frac{dx}{dy} = 0$$

*multiplying by* 3

$$\frac{dx^2}{d^2y} - 12\frac{dx}{dy} = 0$$

We put $\quad x = e^{mt}, \quad \frac{dx}{dt} = me^{mt}, \quad \frac{d^2x}{dt^2} = m^2e^{mt}$

Then the auxiliary equation is

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$$m^2e^{mt} + me^{mt} = 0$$

$$m^2 + 5m = 0$$

$$m(m+5) = 0$$

$$m = 0 \quad , \qquad m + 5 = 0$$

$$m = 0 \quad , \qquad m = -5$$

Therefore, the auxiliary equation has distinct real roots

**Question No 3:** Solve the following intial value problem

$$\int \frac{dT}{T - Tm} = \int K\ (dt), T\ (0) = T\ 0$$


$$\int \frac{dT}{T - Tm} = \int k dt \qquad ,T(0) = 0$$

$$\therefore \int \frac{dT}{t(1-m)}$$

$$\int uvdv = \int u'vfv + \int uv'fv$$

$$\int \frac{dT}{t(1-m)} = \int \frac{1}{t} * \frac{1}{1-m} dt$$

$$\frac{1}{1-m} \int \frac{1}{t} dt = \frac{1}{t} \int \frac{1}{1-m} dt$$

$$\frac{1}{1-m} * \ln t = \frac{1}{t} \frac{1}{1-m} t$$

$$\frac{2}{1-m} * \ln t$$

$$\frac{2}{1-m} * \ln t = k \frac{t^2}{2}$$

$$\ln t = k \frac{t^2}{2} \left( \frac{1-m}{2} + c \right)$$

$$t = e^{k \frac{t^2}{2} \left( \frac{1-m}{2} + c \right)}$$

$$t(0) = 0$$

$$0 = e^{0 \frac{0^2}{2} (0+c)}$$

$$c = 0$$


**Question No 4:** Factorize the differential equation $D^4 - 8d^2 + 16$ if possible

$D^4 - 8d^2 + 16$

$D^4 - 8d^2 + 16 = (D^2-4)(D^2-4)$

$\left(D^4 - 8d^2 + 16\right)y = (D^2-4)(D^2-4)y$

$(D^4 - 8D^2 + 16)y = (D+2)(D-2)(D+2)(D-2)y$

$(D^4 - 8D^2 + 16)y = (D+2)y(D-2)y(D+2)y(D-2)y$

*To verify* this we let

$w=(D+2)y=y'+2y$

$(D-2)w=D(y'+2y)-2(y'+2y)$

$(D-2)w = y''+2y'-2y'-4y$

$(D-2)w = y''-4y...........eq(1)$

*so* $w=(D+2)y$

$(D-2)(D+2)y = y''-4y$

$w=(D-2)y = y'-2y$

$(D+2)w = D(y'-2y)+2(y'-2y)$

$(D+2)w = y''-2y'+2y'+4y$

$(D+2)w = y''+4y$

*so* $w=(D-2)y$

$(D+2)(D-2)y=y''+4y..........eq(2)$

therefore we can write from the two expressions

$(D-2)(D+2)y = (D+2)(D-2)y$

**Question No 5:** Find the wronsktin of the differential equation

Y``` -2y``-18y = 6 + $4^{-t}$ using variation of parameter and the root of the auxiliary equation is m1 = -3 , m2 = -1 , m3 = 6?


**Question No 6:** A radioactive isotope has a half-life of 16 days we have 30 g at the end of 30days, how much radio isotope was initially present?

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